Artificial Intelligence

Lessons 19–23 Active Learning Study Guide

Instructions: Use the following questions to guide your studying. Rather than memorizing definitions, focus on explaining the underlying concepts, performing the mathematical reasoning, and applying these probabilistic theories to practical scenarios.

Lesson 19 — Bayes' Rule

19.1 Explain the mathematical components of Bayes' Rule. How does the prior probability interact with the likelihood of new evidence to produce an updated posterior probability?

19.2 What is the "base rate fallacy"? Provide a concrete example scenario (e.g., medical testing or spam filtering) where ignoring the base rate leads to a drastically incorrect interpretation of evidence.

19.3 Given a scenario where an agent receives a piece of evidence, walk through the mathematical steps to apply Bayes' Rule. What does the resulting posterior probability physically represent regarding the agent's new state of belief?

Lesson 20 — Bayesian Networks

20.1 How do the nodes and directed edges in a Bayesian Network graphically represent a complex joint probability distribution? What information is stored within each individual node?

20.2 Define conditional independence and explain how it differs from absolute (marginal) independence. How is conditional independence visually encoded in the structure of a Bayesian Network?

20.3 Interpret common network structures (such as causal chains, common causes, and common effects). How does observing a variable in the middle of a chain alter the probabilistic relationship between the variables on either end?

Lesson 21 — Bayesian Inference

21.1 Explain how the factorization property of Bayesian Networks simplifies the calculation of the full joint probability distribution. Why is this computationally vital for large systems?

21.2 Walk through the process of exact probabilistic inference by enumeration. How do you compute the probability of a query variable given specific evidence by summing out (marginalizing) the hidden variables?

21.3 Once you have computed an inference result, how do you interpret this value in the context of the environment? How should an intelligent agent alter its decision-making based on this calculated probability?

Lesson 22 — Markov Models

22.1 What is the fundamental "Markov Assumption" (or Markov Property) regarding state transitions? Why is this assumption strictly necessary to model dynamic systems over long periods of time?

22.2 How do you mathematically represent a system's dynamics using a state space and a transition matrix? What must be true about the rows of a valid transition matrix?

22.3 If you are given the initial state distribution of a system, explain the mathematical procedure required to compute the probability distribution of the system's state exactly k time steps into the future.

Lesson 23 — Hidden Markov Models (HMMs)

23.1 What is the structural distinction between a standard Markov Model and a Hidden Markov Model (HMM)? Clearly define the distinct roles of the transition model and the sensor (emission) model.

23.2 In a filtering or monitoring task, explain the two-step process of updating beliefs over time. How does the model first project the current state forward, and then how does it use a new observation to update the belief about that hidden state?

23.3 Given a simple sequence of consecutive observations (e.g., an agent observing a robot's sensor readings over three days), how do you conceptually use the structure of an HMM to infer the most likely underlying sequence of actual system states?